

PREM VISHNOI

Data Engineering | Head of Data & AI | Banking & Enterprise Platforms

Data Architecture · Data Integration · Big Data · ETL & Lakehouse · Analytics & AI/ML · 15+ Years

Singapore | Vietnam TRC Holder (Ready to Relocate to Vietnam)

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PROFESSIONAL SUMMARY

Data Engineering leader with 15+ years designing and delivering enterprise data architectures, pipelines, and platforms across banking (Standard Chartered, DBS, PayPal), fintech (Xendit), and large-scale e-commerce/retail (Lazada/Alibaba, CP Group). 6+ years in senior managerial roles leading multi-country data teams of 25-30 engineers.

Deep expertise across the full data engineering stack: dimensional data modelling, ETL/ELT processes, data warehouse and lakehouse architectures, big data technologies (Hadoop, Spark, Flink, Kafka, Airflow), database systems (SQL, NoSQL, Graph), and programming (Python, Scala, Java, R). Proven mastery in Agile team leadership and mentoring, with a strong track record of driving successful digital transformation initiatives at enterprise scale.

Strong grounding in Information Security, data governance, and compliance (MAS, PDPA, GDPR, AML) from Tier-1 banking environments. Regularly translate complex data architecture and business requirements into scalable data solutions, partnering with Data Scientists, Business, Finance, Risk, and IT stakeholders up to CEO / CIO / CDO level.

CORE STRENGTHS

- **Data Architecture:** Enterprise data strategy, dimensional modelling, data warehouse & lakehouse design, real-time & batch architectures, evaluation and selection of emerging data technologies, best-practice frameworks for scalable data infrastructure.
- **Data Integration & Pipelines:** Design and delivery of ETL / ELT pipelines from diverse sources (transactional, streaming, third-party), optimised for reliability, quality, and reusability — enabling BI, data science, ML, and real-time analytics platforms.
- **Analytics & AI Enablement:** Partner with Data Scientists to translate data requirements into reusable data assets and feature-ready datasets that accelerate ML model build, deployment, and monitoring in production.
- **Project & Delivery Management:** Manage delivery of complex data programmes across dynamic business requirements — resolve project roadblocks, drive RCA and post-mortems, ensure delivery in Agile teams with high performance and predictability.
- **People Management:** Attract, onboard, and retain top data engineering talent; scaled teams from 0 → 25-30; set KPIs / KRAs, run performance cycles, mentor engineers to senior levels; role-model culture, coaching, and continuous learning.

KEY SKILLS

- **Data Architecture & Modelling:** Dimensional Modelling (Star/Snowflake), Data Warehouse, Data Lakehouse (Delta Lake, ODS/CDM/ADS), Enterprise Data Strategy, Real-Time Analytics, Master Data Management
- **Big Data Technologies:** Apache Spark, Kafka, Apache Flink, Airflow, Hadoop, Hive, dbt, ClickHouse, HBase, Databricks, Teradata
- **Databases:** SQL (Oracle, MSSQL, PostgreSQL, MySQL), NoSQL (MongoDB, HBase, DynamoDB), Data Warehouse (Teradata, Snowflake exposure), Graph Databases (exposure)
- **Programming Languages:** Python, SQL, Scala, Java, R (exposure), PySpark, Shell / Bash, Git, CI/CD
- **Cloud & Infrastructure:** AWS, Azure, Alibaba Cloud, Databricks, Kubernetes, Docker, Terraform, Multi-Cloud Architecture, Cost Optimisation
- **AI / ML Engineering:** End-to-End ML Lifecycle, MLOps, MLflow, Pricing, Segmentation, Recommendation, Forecasting, AML/Anomaly Detection, GenAI (LLMs, RAG, Text-to-SQL)
- **Data Governance & Security:** Data Quality, Metadata Management, Data Lineage, Data Contracts, Information Security, Access Controls, Compliance (MAS, GDPR, PDPA, AML)
- **Delivery & Leadership:** Agile / Scrum, Team Scaling (0→30), KPI/KRA/OKR Management, Budget Ownership (\$3M+), Vendor Management, Stakeholder Alignment, Digital Transformation
- **Analytics:** Power BI, Tableau, Executive Dashboards, Self-Service BI, Business Intelligence Strategy, Data Mining

EXPERIENCE

Head of Data Engineering

Makro (CP Group) | July 2024 – Present

Lead the data, analytics, and AI platform strategy for an \$8B omnichannel business across 3 countries — own architecture, delivery, \$3M budget, vendor management, and a 30-person organisation spanning Data Engineering, Analytics, BI, and AI.

- **Data Architecture:** Defined and executed the enterprise data strategy, modernising legacy systems into a Databricks Lakehouse processing 10B+ rows daily across finance, supply chain, retail, and e-commerce domains — established best-practice architecture patterns and reusable components across the bank-like data platform.
- **Dimensional Data Modelling:** Designed dimensional models and Master Data Management structures for finance, customer, supply chain, and O2O domains — enabling consistent reporting, self-service analytics, and downstream ML use cases.
- **Data Integration & Pipelines:** Built optimised ETL / ELT pipelines integrating data from 50+ sources (S3, MongoDB, RDBMS, REST APIs) into a unified gold-standard layer — powering BI, ML, and real-time analytics at 99.99% platform uptime.
- **Analytics Foundation:** Automated month-end financial close from 5 days to 2 hours; delivered 50+ executive dashboards covering revenue, profitability, and O2O performance for CEO / CFO / COO-level decisioning.
- **Data Scientist Enablement:** Partnered with data scientists to build reusable feature datasets and governed data assets that accelerated ML model deployment for pricing, forecasting, segmentation, and item matching.
- **AI & GenAI Delivery:** End-to-end delivery of production AI systems — Dynamic Pricing Model, Chat-with-Data (Text-to-SQL LLM), Item Matching (NLP + embeddings), MarTech Automation, and AI strategy across customer service, content, and search.
- **Project & Delivery Management:** Manage delivery of complex, cross-functional data initiatives — reduced production incident volume by 60% through structured RCA, post-mortems, monitoring, and Agile discipline.
- **People Management:** Scaled the team from 8 to 30 across four domain squads (Finance, Supply Chain, O2O, Platform); set KPIs / KRAs tied to business outcomes; mentored engineers to senior/principal levels.
- **Cost & Governance:** Reduced AWS/Databricks spend by 40% while establishing data governance covering access controls, secrets, lineage, and compliance (GDPR, PDPA) across 3 countries.

VP, Data Engineering

Lazada Group (Alibaba), Singapore | July 2018 – January 2024

Led data platform architecture and strategy for SEA's largest e-commerce business — full data lifecycle from ingestion through analytics, data products, and ML. Earned double promotion in recognition of platform impact and cross-functional leadership.

- **Data Architecture at Scale:** Architected cloud-native Data Lakehouse (ODS / CDM / ADS dimensional layers) enabling 10x data growth and 50% scalability improvement — single source of truth across 600+ governed production tables for BI, data science, and real-time analytics.
- **Real-Time Data Integration:** Built real-time data platform (Kafka, Flink, ClickHouse, HBase) processing 100M+ daily events — powering last-mile logistics analytics engine generating 20M+ daily insights.
- **ML Enablement:** Engineered governed, feature-ready datasets and reusable feature pipelines enabling ML use cases across delivery optimisation, forecasting, supplier performance, and decision support.
- **Production ML Delivery:** End-to-end ownership of ML systems — Logistics Route & ETA Optimisation, Customer Segmentation for Warehouse Fulfilment, with monitoring, retraining, and governance in production.
- **Agile Leadership:** Led multi-country Agile squads across Singapore, Indonesia, Philippines — mentored engineers on Agile principles, Scrum practices, and data platform delivery patterns.

Big Data Consultant — Banking & AML Compliance

Standard Chartered Bank, Singapore (Optimum) | February 2016 – July 2018

Built large-scale AML and compliance data platforms for a Tier-1 global bank across 15+ countries — deep grounding in regulated banking delivery, Information Security, and audit-grade data pipelines.

- **Banking Data Pipelines:** Built Hadoop-based AML compliance pipelines processing transactions across 15+ countries with strict SLAs and audit-grade lineage — supporting MAS regulatory requirements.

- **Information Security & Compliance:** Embedded access controls, data classification, and audit logging into engineering practice; delivered against regulatory audit and compliance reviews — direct exposure to banking Information Security principles.
- **AML & Entity Resolution:** Owned end-to-end AML analytics workflow — feature engineering, suspicious-pattern detection, and Lucid Search fuzzy-matching for high-risk entity screening across multilingual customer records.

Principal Data Engineer — Fintech

Xendit.co, Singapore | January 2024 – June 2024

- Designed Kafka-based streaming architecture for real-time payment transaction processing; refactored dbt and Spark pipelines cutting execution time by 60% and improving payment settlement SLAs.
- Orchestrated FinOps initiative reducing AWS/Databricks spend by 40% while handling growing transaction volumes; led 4-engineer team on batch and real-time payment pipelines.

EARLIER EXPERIENCE

Big Data Developer • DBS Bank (Apar), Singapore | Sep 2015 – Mar 2016 — Consumer Banking data integration, dimensional models in Hive, SCD2 frameworks, real-time ingestion from IBM MQ

ETL Data Engineer • PayPal, India | Apr 2014 – Sep 2015 — Global payments data warehouse integration, ETL migration to Hadoop/Hive, Agile production delivery (Hackathon Winner)

Data Warehouse & Software Engineer • Exilant Technologies (Apple Inc), India | Jan 2011 – Apr 2014 — Enterprise data warehouse solutions, dimensional modelling, ETL automation across core and semantic layers

→ 6+ years of foundational banking, payments, and enterprise data warehouse experience.

EDUCATION

M.S. in Data Science (In Progress) • Northwestern University, USA | Sep 2025 – Jan 2027

Post Graduate Program in Artificial Intelligence and Machine Learning • UT Austin, USA | Jan 2019 – Jan 2020

B.E. Electronics & Communication • University of Rajasthan, India | Jul 2005 – Jul 2009

PROFESSIONAL CERTIFICATIONS

- **Agentic AI Systems & Optimisation** • CMKL Professional Training Centre | Dec 2025
- **AWS Solutions Architect Associate** • AWS | Dec 2023
- Generative AI • Educative

ADDITIONAL INFORMATION

Location & Work Authorisation: Vietnam TRC Holder — ready to relocate to Ho Chi Minh City / Hanoi. Extensive experience across Singapore, Thailand, and Southeast Asia.

Language: English (Professional Fluency) | Hindi (Native)

Recognition: PayPal Hackathon Winner | Double Promotion at Lazada Group (Alibaba)

Thought Leadership: Writer on Medium covering data architecture, AI, and engineering leadership